Alibaba-Zhejiang University Joint Institute of Frontier Technologies

Microsoft Visual Perception Laboratory of Zhejiang University

Research Summary

Yongcheng Jing

College of Computer Science and Technology

Zhejiang University

Hangzhou, China

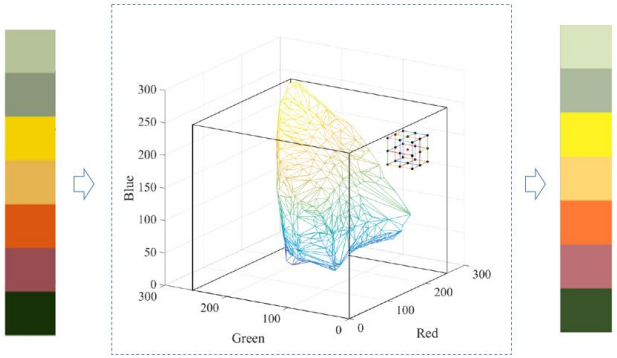
Outline

- Derivations
- Research with Publications
  - Image Style Transfer
  - Human Parsing
  - Fashion Outfit
- Other Research Topics
- Research in Practice

Derivations

- The topic of my research derives from a project **“Intelligent Design for Textile Production”** , supported by *National Natural Science Foundation of China.*

Image Style Transfer



Color Gamut Mapping

Human Parsing and Fashion Outfit

ECCV’18

For Designers

ICME’17

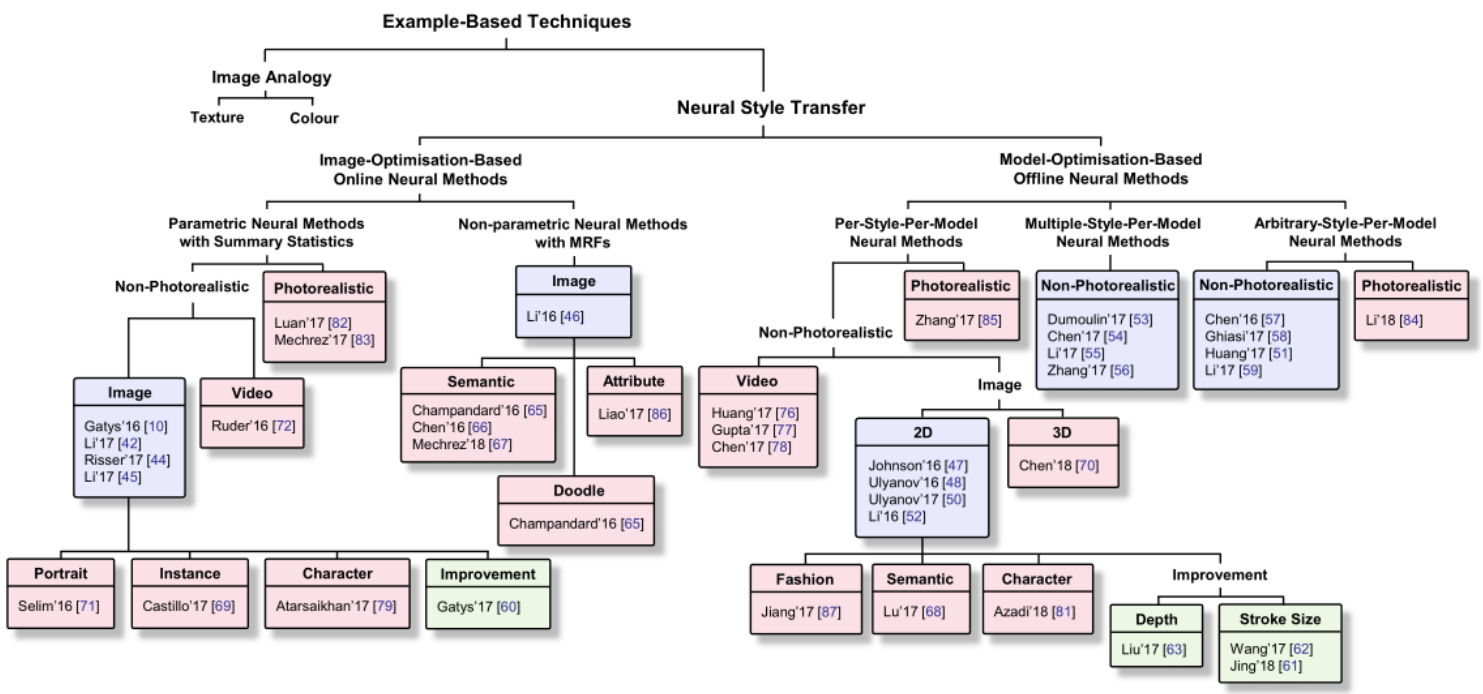
For Engineers

ICME’18 & ICMR’18

For Customers

Image Style Transfer

- **TVCG** Manuscript: **"Neural Style Transfer: A Review"** (**Minor Revision**)
  - The **first comprehensive survey** in the field of Neural Style Transfer.
  - The project has received over **900 stars** on Github.



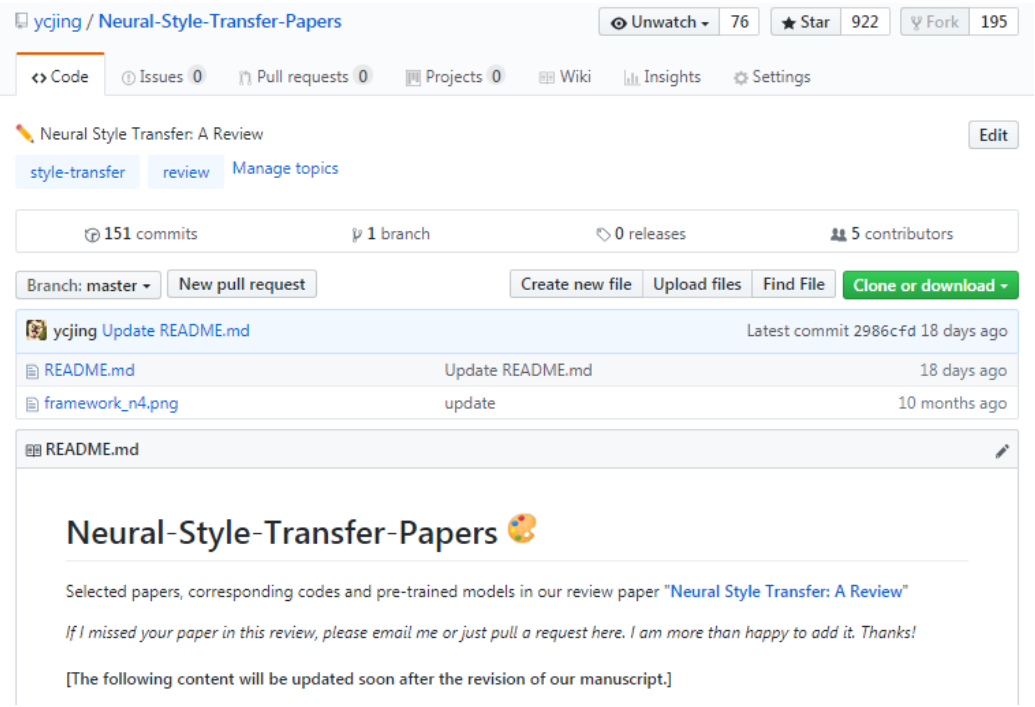


Image Style Transfer

- **ECCV** 2018: **"Stroke Controllable Fast Style Transfer with Adaptive Receptive Fields"**
  - Analyze the **factors** that influence the stroke size in style transfer.
  - Propose the **first style transfer network** that achieves continuous and spatial stroke size control.
  - Present two corresponding **training strategies** to achieve faster convergence and augment new stroke size upon a trained model.

Content & Style | **Stroke Size #1** | **Stroke Size #2** | **Stroke Size #3** | **Mixed Stroke Size**

Demo video: https://youtu.be/m0-ob970Tws

Image Style Transfer

- **ECCV** 2018: **"Stroke Controllable Fast Style Transfer with Adaptive Receptive Fields"**
  - Network Architecture:

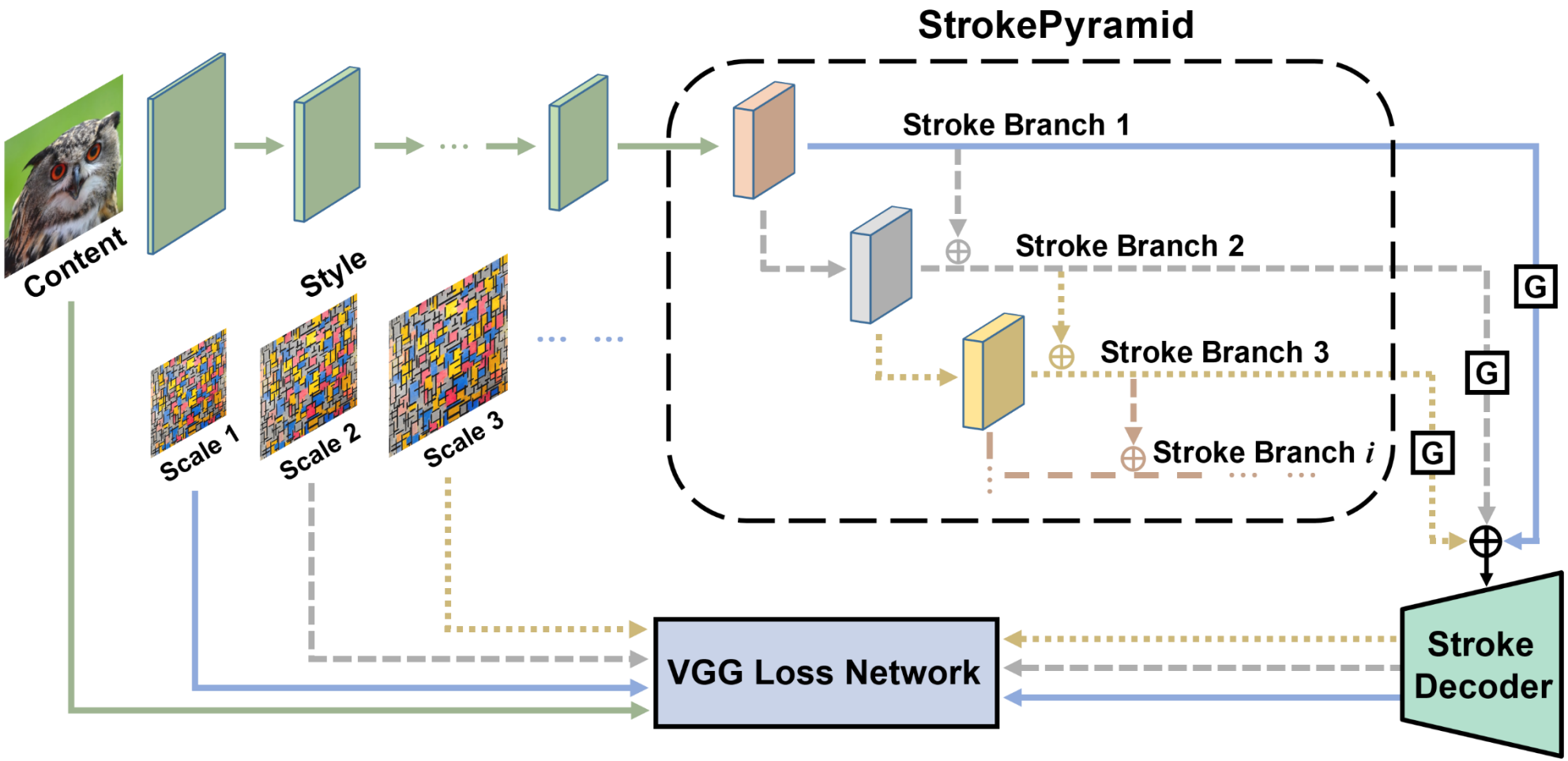
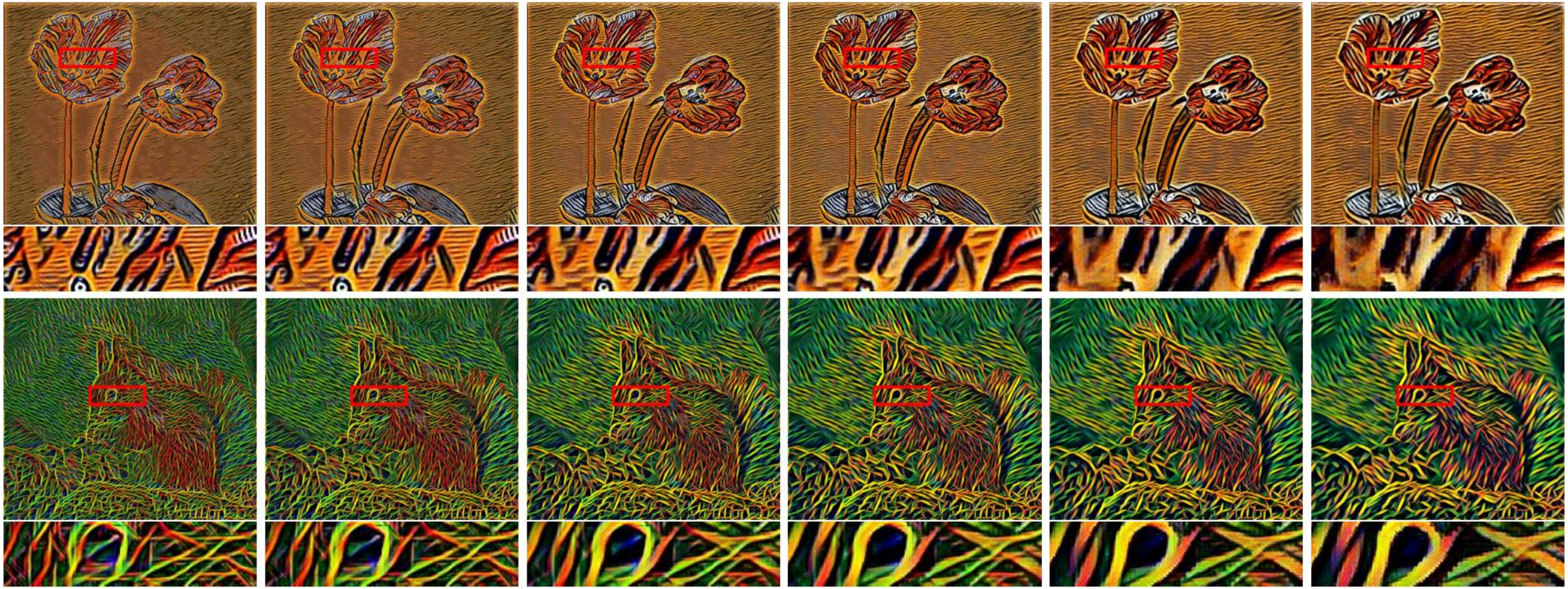


Image Style Transfer

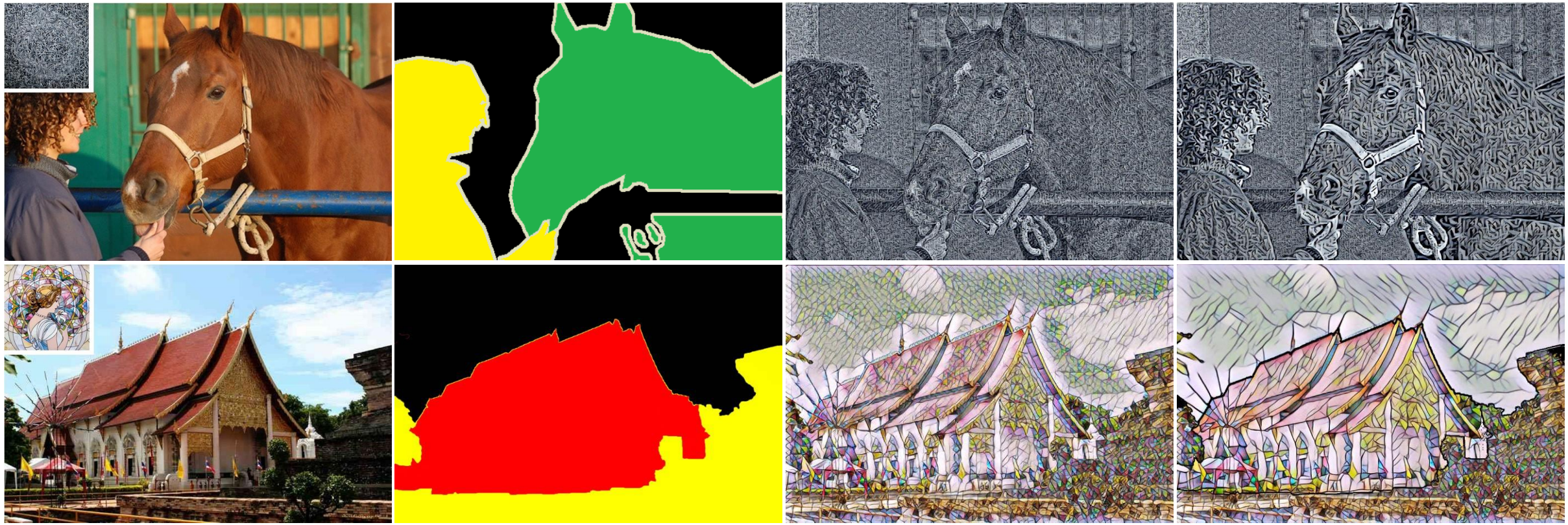
- **ECCV** 2018: **"Stroke Controllable Fast Style Transfer with Adaptive Receptive Fields"**
  - Results of **Continuous Stroke Size Control:**



*** We zoom in on the same region (red frame) to observe the variations of stroke size**

Image Style Transfer

- **ECCV** 2018: **"Stroke Controllable Fast Style Transfer with Adaptive Receptive Fields"**
  - Results of **Spatial Stroke Size Control:**



Content & Style | **Semantic Segmentation** | **Same Stroke Size** | **Spatial Stroke Size Control**

Image Style Transfer

- **Project: "Color Style Transfer"**
  - By a summer research group of undergraduate students led by me.
  - The first TensorFlow Implementation of a color style transfer algorithm.
  - The project has received over **500 stars** on Github*.

Content

Color Style

Result

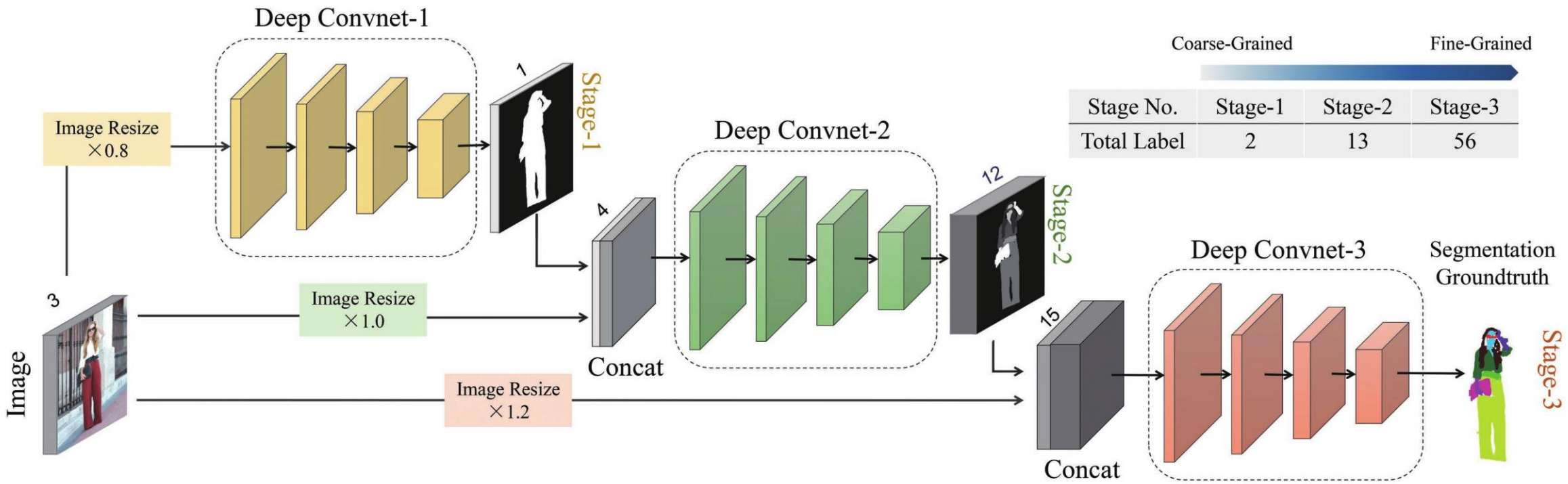
Acknowledgement

- This work was done when Yang Liu was a research intern at *Alibaba-Zhejiang University Joint Research Institute of Frontier Technologies*, under the supervision of Prof. Mingli Song and Yongcheng Jing.

***Github: https://github.com/LouieYang/deep-photo-styletransfer-tf**

Human Parsing

- **ICME** 2018: **"Finer-Net: Cascaded Human Parsing with Hierarchical Granularity"**
  - Propose a cascaded human parsing architecture with hierarchical coarse-to-fine granularity.
  - Network Architecture:



Coarse-Grained → Fine-Grained

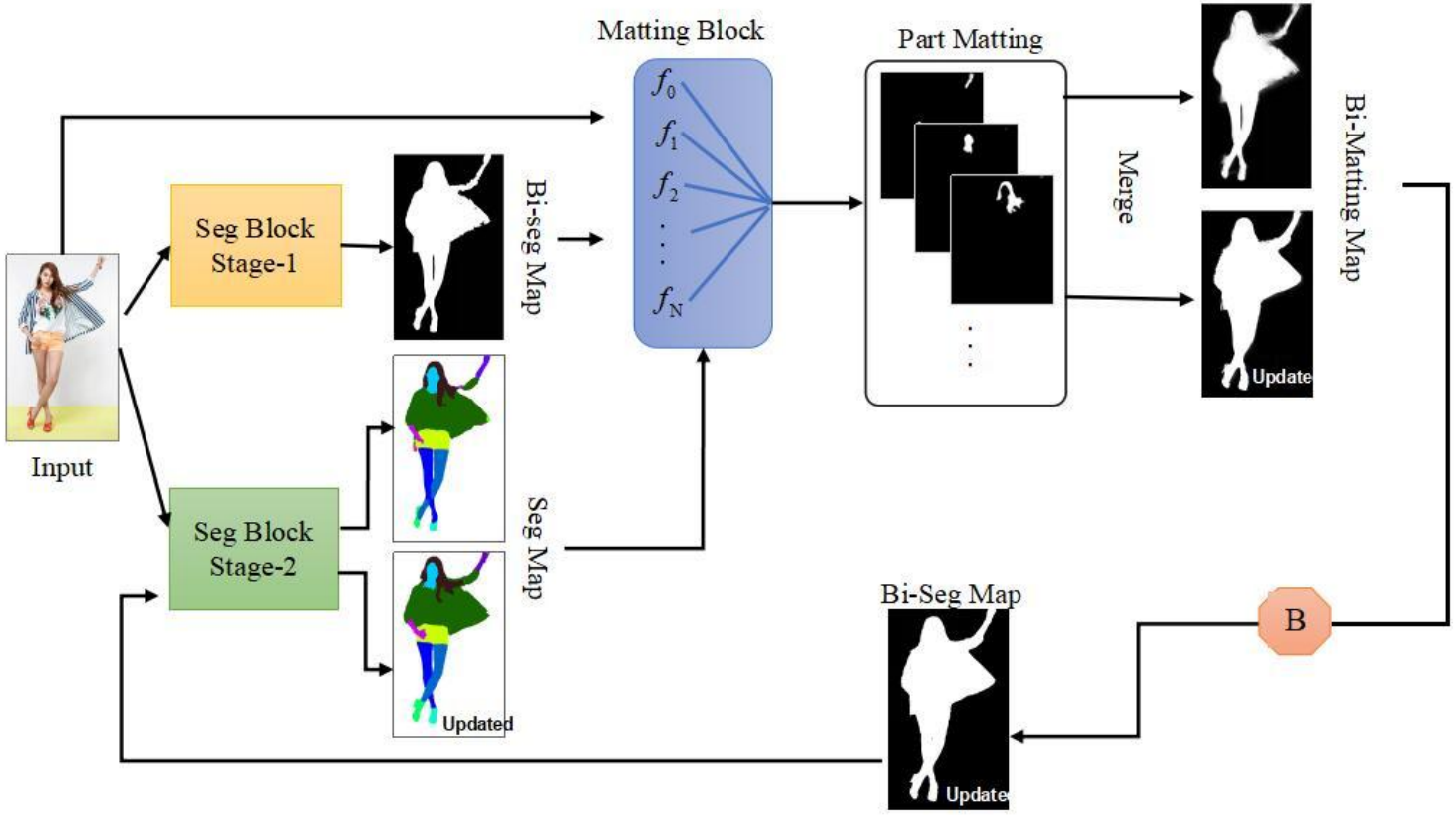
Stage No.	Stage-1	Stage-2	Stage-3
Total Label	2	13	56

Human Parsing

- **TIP** submission (**Major Revision**):

  "Edge-Sensitive Human Cutout With Hierarchical Granularity and Loopy Matting Guidance"

  - Study a hybrid model of joint parsing and matting: **1)** Achieve the state-of-art parsing performance on the standard human-parsing benchmark. **2)** Address the region-specific matting problem in an unsupervised manner.
  - Network Architecture:

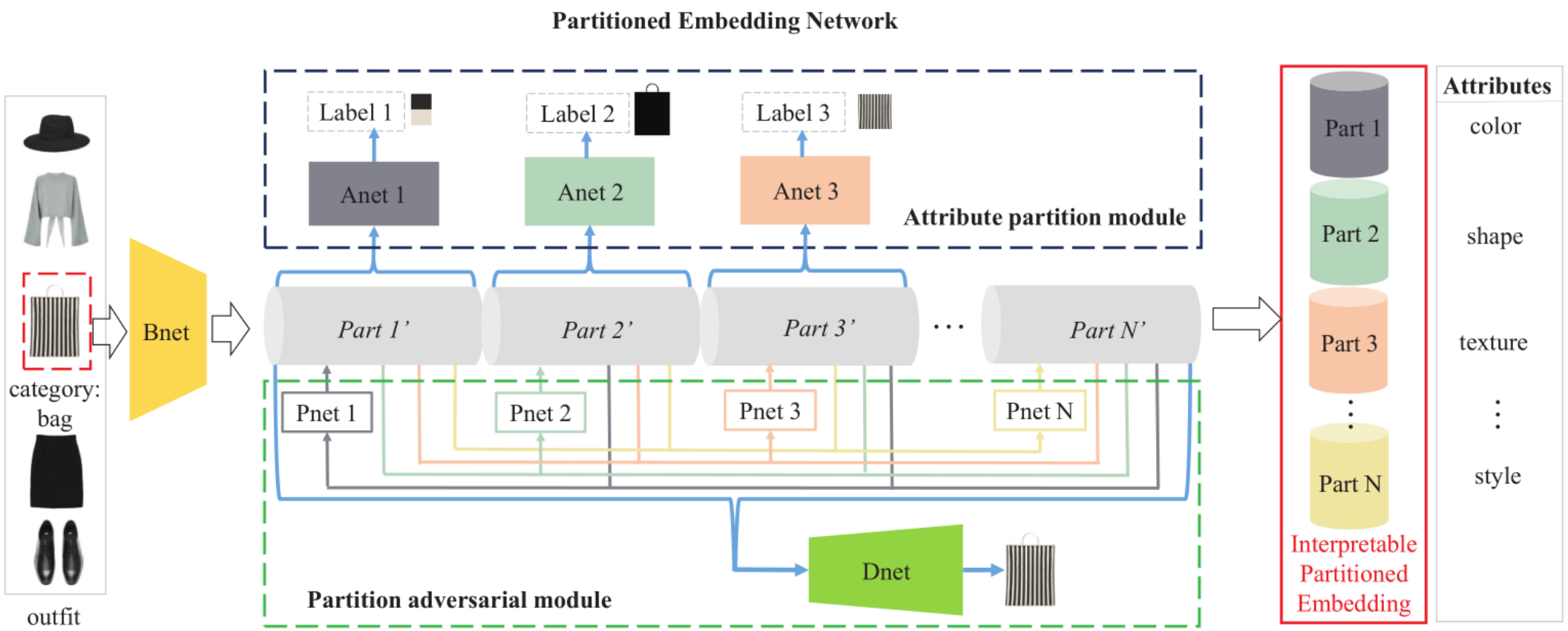


Fashion Outfit

- **ICMR** 2018:

  "Interpretable Partitioned Embedding for Customized Multi-item Fashion Outfit Composition"

  - Network Architecture:

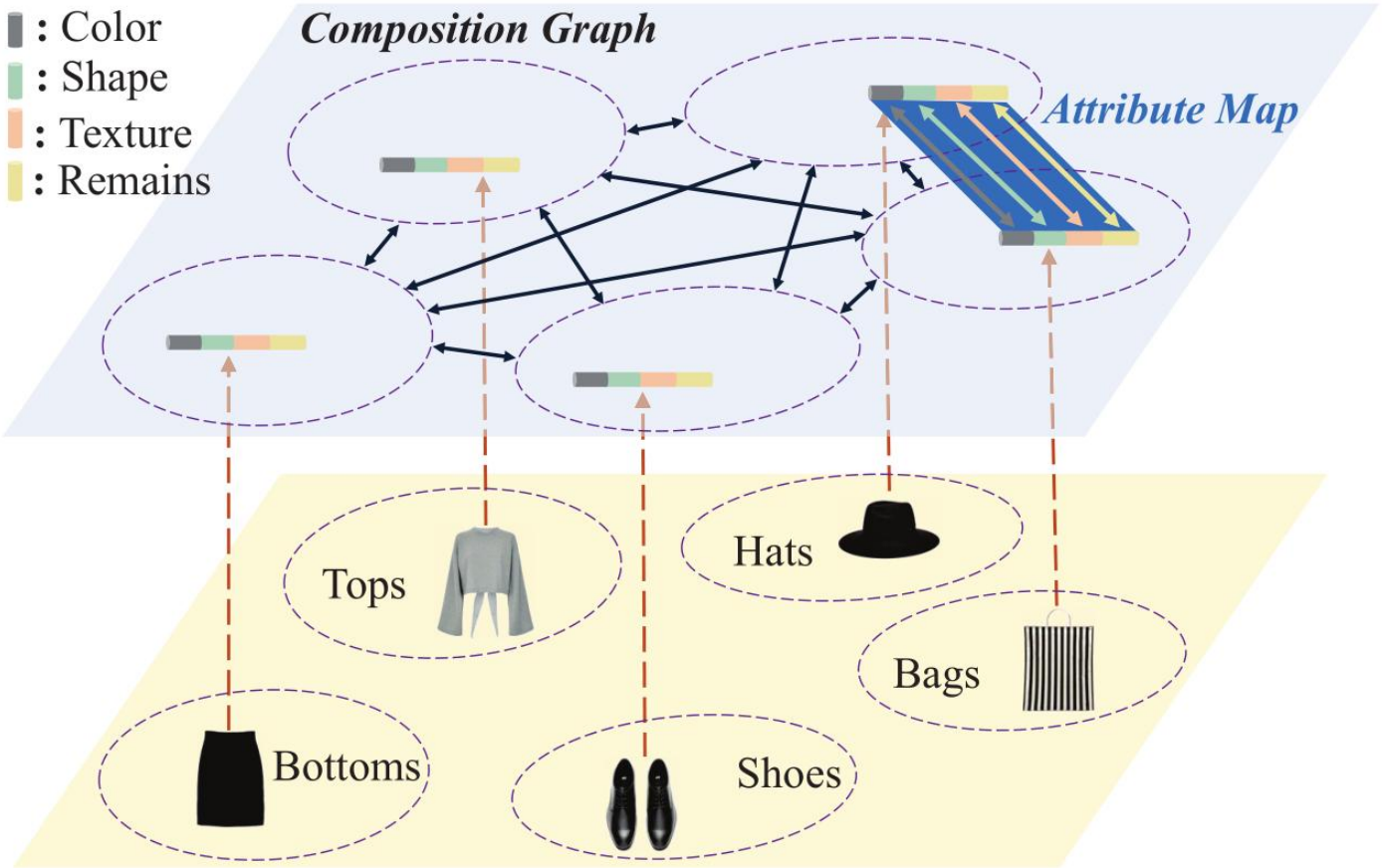


Fashion Outfit

- **ICMR** 2018:

  “Interpretable Partitioned Embedding for Customized Multi-item Fashion Outfit Composition”

  - Fashion Outfit Composition Graph:



Other Research Topics

- Also have some **knowledge** in the following areas:
  - Network Visualization
  - Generative Adversarial Networks
  - Adversarial Samples
  - Deep Image Aesthetics Assessment
- **Related Talks***:
  - "*A Review on Adversarial Attacks*" (June 4, 2018)
  - "*Deep Image Aesthetics*" (Jan 17, 2018)
  - "*Generative Adversarial Networks: Recent Advances and Popular Application*" (Mar 20, 2017)
  - "*Visualizing Convolutional Neural Networks: Image Reconstruction*" (Oct 17, 2016)
- **Current Project:**
  - Turing Attackers into Guards: Adaptive Image Steganography via Adversarial Examples
  - Submitted to ICCV 2019

Research in Practice

- **Fashion Outfit:** Recommending products for customers.

- Online on **Taobao.com** during **11.11. & 12.12. Shopping Festival**.
- Involving **two million** outfits.
- **CTR:** ~50%

The work is done during my internship in Taobao AI Team of Alibaba Group.

Research in Practice

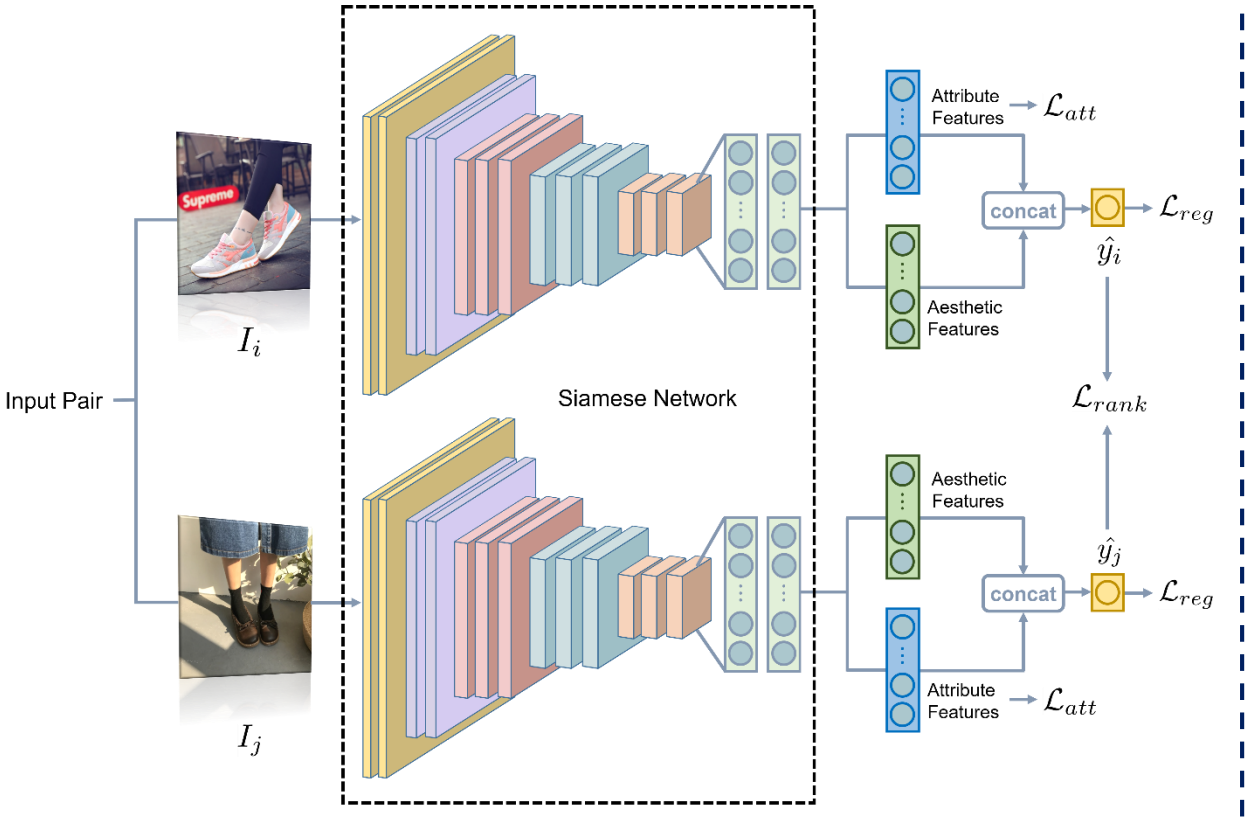
- **Human Parsing + Image Style Transfer** = Automatic Portrait Fusion

- Page View (PV): **339, 187** (1 day)
- User View (UV): **82, 210** (1 day)
- Usage Count: **3** (per user)

The work is done during my internship in Taobao AI Team of Alibaba Group.

Research in Practice

- **Product Image Aesthetics Assessment:** For product image ranking and data cleaning.
  - Implemented an image aesthetics assessment algorithm and collected a dataset by hiring some workers.
  - The conclusion from this project: In China, product image aesthetics **have little to do** with sales.



Bad:

Good:

The work is done during my internship in Taobao AI Team of Alibaba Group.

Thanks！